



# Temperature and Flow Measurements in Incompressible Heated Jets

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## Abstract

**An experimental study is conducted to perform time-resolved temperature and velocity measurements on a Mach 0.08 jet at total temperatures of 295 K and 353 K. Mean and rms temperature data acquired using two fine wire sensors with diameters 1.3 and 3.8  $\mu\text{m}$  are compared. In order to extend the limited frequency response of the wires, the temperature data are post-processed using a frequency compensation technique available in the literature. Corresponding velocity measurements are performed at cold and heated conditions using single and parallel wire probes. Simultaneously measured temperature and velocity data obtained using the parallel wire probe are used to calculate correlations pertinent to axial turbulent heat flux. In addition to shedding some light on the aerothermal properties of heated turbulent jets, vis-à-vis their cold counterparts, this study also provides a database for numerical prediction of these flows.**

## I. Introduction

Flow physics of high-temperature turbulent jets is of great interest for the aeroacoustics and propulsion communities. Since the advent of jet engines, a large amount of research effort has been dedicated towards measurements of turbulence statistics in jets to better understand their aerothermal and aeroacoustic properties. Accurate measurement of instantaneous quantities allows calculation of higher-order turbulence statistics and space-time correlations that are necessary for detailed understanding, for example, of the hydrodynamic near field and its relation to the acoustic far-field. In addition, establishment of a consensus turbulence database is critical for development of numerical models for plume and noise prediction. A reliable estimation of turbulent heat flux terms of the Reynolds-averaged energy equation necessitates simultaneous measurement of temperature and velocity components. Technological advancement in the last few decades has helped mature a number of velocity measurement techniques such as Hotwire Anemometry (HWA) [1, 2], Laser Doppler Velocimetry (LDV) [3, 4], and Particle Image Velocimetry (PIV) [5, 6], providing accurate mean and instantaneous details of the jet velocity field. However, reliable measurement of time-resolved temperature in turbulent flows remains a challenge. A number of studies have explored the use of cold wire sensors 1-5  $\mu\text{m}$  in diameter for measurement of time-resolved temperatures in low-speed turbulent flows [7, 8, 9]. Heat flux probes consisting of velocity and temperature sensors in close proximity, for example an X-wire offset from a cold wire sensor, have also been used to study turbulent transport of heat in various flow configurations [10, 11, 12]. In addition to their intrusive nature, one of the major shortcomings of the fine wire sensors is their low frequency response. Depending on the wire thickness, the cut-off frequencies can range from a few hundred Hz to a few kHz. This poses serious limitations for temperature measurements in turbulent flows, especially at high speeds. Optical techniques that utilize laser-based molecular scattering of light, such as Coherent Anti-Stokes Raman Spectroscopy (CARS), Laser Induced Fluorescence (LIF), Raman and Rayleigh scattering, have also been investigated [13, 14, 15]. While these methods have the advantage of non-intrusive measurement with high spatial resolution, they are limited by low-repetition rates of commercially available lasers. In most cases these techniques leverage short pulse duration of the lasers (3-5 ns) and averages over large ensembles to obtain mean and rms temperature data. In addition, optical techniques are often limited by weak scattered signals and burdened by the complexity of hardware and painstaking optical alignments. Due to these limitations, the literature on time-resolved temperature measurement is very limited. This study utilizes the fine wire sensor technique and, with an eye on its limitations, explores the properties of a low-speed heated jet. A brief description of the technique is discussed next.

### A. Temperature measurement using fine wire sensors

The principle of operation of a fine wire temperature sensor is simple. A small constant current ( $I < 1$  mA) is supplied to the wire operating it at a low Overheat Ratio (OHR). At low OHR, the wire's sensitivity to velocity is negligible

and the change in wire's resistance is proportional to the changes in flow temperature [16]. It is now well known that the dynamic response of the fine wire sensors are governed by two main factors, thermal inertia of the wire and losses due to end conduction. These are briefly described in the following.

Equation 1 shows the relationship between the resistance of the wire and its temperature. Here,  $R_w$  is the wire resistance at temperature  $T_w$ ,  $R_{ref}$  is wire resistance at reference temperature  $T_{ref}$  and  $\alpha$  is the temperature coefficient of resistance.

$$R_w = R_{ref}(1 + \alpha(T_w - T_{ref})) \quad (1)$$

At a low OHR, the 'cold' wire assumes the temperature of the local flow such that  $T_w = T_a$ . In principle, the resistance of the wire in time ( $t$ ) changes with the temperature of the local flow as shown in eqn. 2.

$$R_w(t) = R_{ref}(1 + \alpha(T_a(t) - T_{ref})) \quad (2)$$

However, this relationship only holds for an ideal, infinitely thin wire that can instantaneously respond to temperature fluctuations in the flow field. Most practical wires have diameters ( $d_w$ ) between 1 and 5  $\mu m$ , and therefore suffer lag in response due to thermal inertia. Hence, it is a common practice to model wire response to temperature fluctuations as a first order system with a time constant. Convective heat balance for a long wire element with negligible end losses can be written as shown in Eqn. 3 [16]. Here,  $m_w$  is the mass of the wire and  $c_w$  its specific heat capacity,  $A$ ,  $B$  and  $n$  calibration constants,  $U$  the flow velocity,  $R_a$  the resistance of the wire at ambient temperature, and  $I$  the current passing through the wire.

$$\frac{m_w c_w}{\alpha R_{ref}(A + BU^n - I^2)} \frac{dR_w}{dt} + R_w = \frac{A + BU^n}{A + BU^n - I^2} R_a \quad (3)$$

This is a first order equation for the wire response with time constant  $\tau$ ,

$$\tau = \frac{m_w c_w}{\alpha R_{ref}} \frac{1}{(A + BU^n - I^2)}, \quad (4)$$

which can be further expressed in the form of Nusselt number (Nu) with  $k$  being the thermal conductivity of the wire [16].

$$\tau = \frac{\rho_w c_w d_w^2}{4kNu} \frac{R_w}{R_a} \approx \frac{\rho_w c_w d_w^2}{4kNu} \quad (5)$$

The equation shows that the frequency response depends not only on the heat capacity of the wire but also on the local flow velocity. Sensor cut-off frequencies can be extended by choosing wires with very fine diameters ( $d_w < 1 \mu m$ ). The result, however, is an extremely fragile sensor, a trade-off that renders it impractical for use in most applications. The range of frequencies where the thermal lag due to the wire is not significant can be estimated using -3dB cutoff frequency given by  $f_{-3dB} = \frac{1}{2\pi\tau}$ .

End conduction losses, on the other hand, occur as a result of the difference in dynamic responses between the sensing wire and the support prongs. The prongs have much larger heat capacity compared to the wire, and therefore are not capable of resolving temperature fluctuations in the flow, except at very low frequencies. Due to this the temperature of the prongs remains approximately equal to the mean temperature of the flow. This results in conduction along the wire leading to heat losses to the prongs [17, 18, 19, 20]. Unlike the effect of thermal inertia, losses due to end conduction occur at low to moderate frequencies. The effect is manifested in the amplitude curve of the frequency response function ( $H(\omega)$ ) as a step reduction in gain occurring near the cut-off frequency dictated by the prong's time constant,  $\tau_p$ , where  $\tau_p \gg \tau$ . A sample frequency response curve for one of the wires used in the current study is calculated using the theoretical method of Tagawa et al. [21] and is shown in fig. 1. Even though the frequency response remains flat for a range of low to mid frequencies, the attenuation in amplitudes can be a sources of significant errors. The attenuation factor,  $H_p$ , can be estimated using the following relation [16].

$$H_p = 1 - 2l_c/l_w \quad \text{where,} \quad l_c = \frac{d_w}{2} \left( \frac{k_w}{k_f} \frac{1}{Nu} \right) \quad (6)$$

Here,  $l_c$  is the cold length which relates to the portion of the wire where the effects of support prongs are felt,  $l_w$  the length of the wire, and  $k_f$  is the thermal conductivity of the fluid. End conduction effects can be minimized by using a wire with large aspect ratio,  $l_w/d_w > 1000$ . However, spatial averaging in longer wires can increase overall errors in measured turbulent quantities [22].

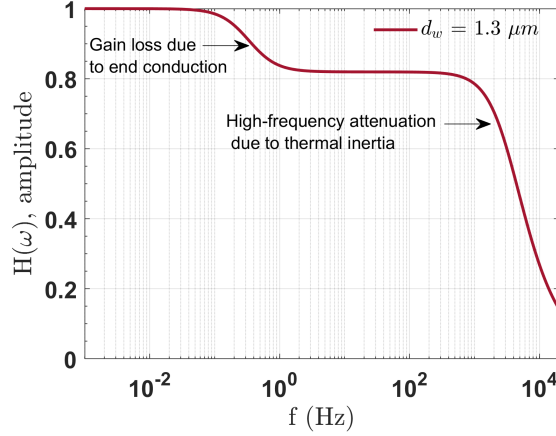


Figure 1: Sample frequency response function of a cold wire showing attenuations due to thermal inertia and end conduction effects.

Thus, in order to assess the accuracy of measured temperature, it is critical to estimate the overall dynamic response of the wires. This involves taking into account not only the effect of thermal inertia but also conduction losses to the support structures. However, due to difficulties associated with measuring time constants of the prongs and wires, most measurements rely on static calibrations where the wire is assumed to be a first order system with flat frequency response. A few studies have addressed the issues theoretically, by solving time-dependent heat transfer equations for the wire and supports, and/or using empirically estimated frequency response function of the wires [23, 20, 21, 24]. Frequency response solution is in the form of a transfer function  $H(\omega)$  that incorporates complex interactions between the wire and its supports (e.g. stubs and prongs). The modulus of  $H(\omega)$ , as shown in fig. 1, reflects signal attenuation at different frequencies. The phase angle (not shown here) represents phase lag between the fluctuations in the flow and the wire's response. In an attempt to restore the attenuated signal, these studies also readily applied appropriate frequency compensation to the raw data and corrected for the response delay of the given wire. The compensation process usually involves performing Fast Fourier Transform (FFT) of the raw time series data and correcting the spectrum for amplitude and phase by multiplying with the inverse of the complex frequency response function ( $H(\omega)^{-1}$ ). An inverse FFT is then applied to recover the compensated temperature time series. This procedure will also be applied to the current data. Using various compensation techniques, studies have demonstrated that more practical cold wire diameters or aspect ratios can potentially be used for temperature measurements in turbulent flows [25].

## B. Scope of the Current Work

The current work presents detailed velocity and temperature measurements in a heated incompressible jet using fine wire sensors. The results in this report are preliminary and involve mean and rms velocity and temperature distributions. In particular, rms temperatures measured using cold wires of two different diameters,  $d_w = 1.3$  and  $3.8 \mu m$ , are compared and differences in the dynamic responses are discussed. Frequency compensation scheme of Tagawa et al. [21] is applied and the response correction is compared using the raw and compensated signals. Brief results for axial turbulent heat flux,  $\overline{u'T'}$ , obtained using a parallel wire probe are presented. In addition to advancing our understanding of thermal mixing in turbulent jets, the findings from the present effort will help evaluate the efficacy of cold wire sensors as a useful tool for measuring temperature fluctuations in heated mixing layers. While the current results are compared to those available in the literature, further direct comparisons will be made in the future using Rayleigh scattering technique in the same rig.

## II. Experimental setup and measurement techniques

### A. Facility and measurement conditions

The experiments are performed in a small heated jet facility in a test cell (ERB CW-13) at NASA's Glenn Research Center. Figure 2a shows the schematic of the facility highlighting various components. Figure 2b shows a picture of the nozzle with a hotwire probe placed downstream of its exit. Compressed air is supplied to a cylindrical plenum 4 ft long and 6 in. in diameter. A 9 kW inline heater is placed upstream of the plenum to heat the incoming flow at desired temperatures. The downstream end of the plenum is fitted with a flange and a 3 in. cylindrical adapter

where various nozzles can be attached. For the current study, flow exits the plenum through a 1 in. converging axisymmetric aluminum nozzle. The internal contour of the nozzle is defined by a third order polynomial. Stagnation conditions in the plenum are monitored using a Setra model 239 differential pressure transducer and a dual sensor T-type thermocouple. In addition, a Setra model 204 absolute pressure transducer and a K-type thermocouple are used to measure ambient conditions during each run.

Measurements are performed on a Mach 0.08 jet at cold ( $T_0 = 295$  K) and heated ( $T_0 = 353$  K) conditions, where  $T_0$  is the total temperature measured in the plenum. The jet Reynolds number based on nozzle exit diameter ( $D$ ) and velocity ( $U_j$ ) at cold and heated conditions are approximately 45000 and 36500, respectively.

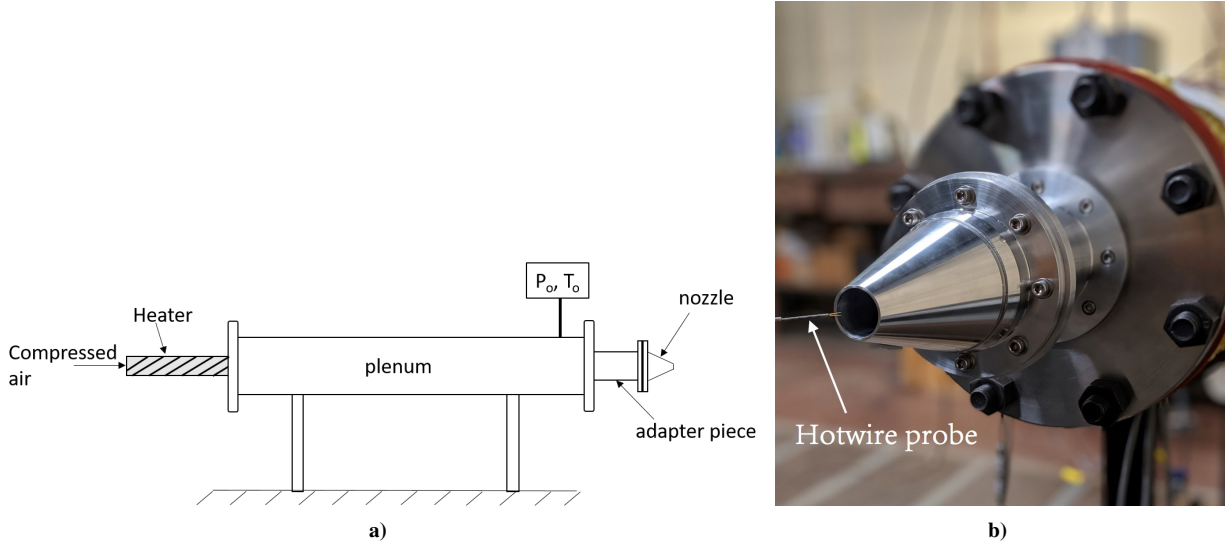


Figure 2: Experimental setup showing (a) a schematic of the CW-13 facility, (b) nozzle with a hotwire sensor downstream of the exit.

## B. Measurement technique

Flow measurements are performed using an IFA-300 thermal anemometry system manufactured by TSI Inc. The system is equipped with four constant temperature (CT) channels for velocity and one constant current (CC) channel for temperature measurements. The constant current channel, also known as the temperature module, supplies a small current ( $< 1$  mA) to the wire keeping it at a low overheat ratio. Bridge voltage values are acquired at each measurement location to determine the changes in wire resistance with temperature. The principle of operation of a cold wire is reviewed in section I.

Temperature measurements are performed using two different single sensor probes, a Dantec 55P31 and a TSI 1260A-T1.5. The 55P31 probe consists of a single platinum wire sensor,  $d_w = 1.3 \mu m$  and  $l_w = 0.4$  mm, suspended between silver plated stainless steel prongs. The wire is directly soldered to the prongs allowing the entire wire length to act as the sensing length ( $l_s$ ). The wire is operated at a low current of 0.5 mA. The extremely fragile nature of this sensor limits maximum flow velocities to about 60 m/s. For the 1260A-T1.5 probe, the T1.5 sensor is a platinum coated tungsten wire with  $d_w = 3.8 \mu m$ ,  $l_w = 1.5$  mm and  $l_s = 1.25$  mm. The wires are soldered directly to gold plated stainless steel prongs, but the ends of the wire are plated with copper to isolate the active sensor region from the prongs. The T1.5 wire is operated with a current of 1 mA. Temperature measurements are performed only for the heated jet at  $T_0 = 353$  K. The -3dB cut-off frequency ( $f_c$ ) due to thermal inertia of the sensors is estimated using eq. (5), where  $f_c = 1/(2\pi\tau)$ . Figure 3 presents cut-off frequencies as a function of mean flow velocity ( $U$ ) for the two sensors. Note that curves for the cold, operating and an intermediate temperature are shown. While a significant effect of wire size on  $f_c$  is evident, the impact of temperature for the range relevant to this experiment is fairly small.

Corresponding axial velocity measurements are performed at cold condition ( $T_0 = 295$  K) using a TSI 1260A-T1.5 single sensor hotwire probe operated in constant temperature mode with an overheat ratio of 1.8. Simultaneous axial velocity and temperature measurements at heated conditions ( $T_0 = 353$  K) are performed using a TSI 1244-T1.5 parallel wire probe. The probe has two T1.5 wires ( $d_w = 3.8 \mu m$ ,  $l_w = 1.5$  mm and  $l_s = 1.25$  mm) that are 0.05 in apart and oriented such that the sensors are aligned perpendicular to the jet axis. One of the wires is operated in CT mode with OHR = 1.8 to measure velocity. The other wire is heated with a 1 mA constant current to perform temperature measurements. A side by side comparison of the physical sizes of the three probes used in the current study are shown in the picture of fig. 4.

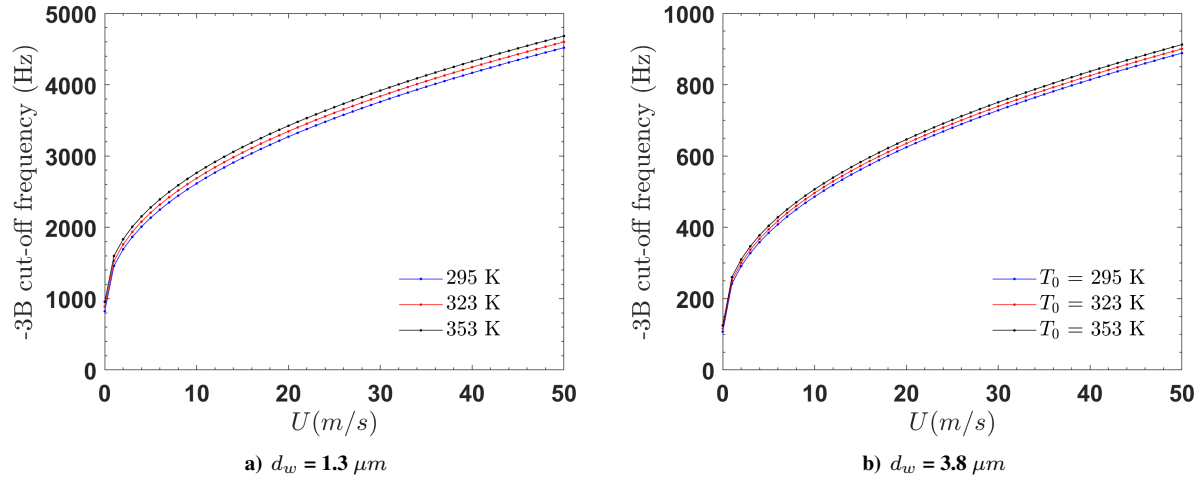


Figure 3: Wire cut-off frequencies as a function of mean flow velocities estimated using eq. (5) for sensors with (a)  $d_w = 1.3 \mu m$  (b)  $d_w = 3.8 \mu m$ .

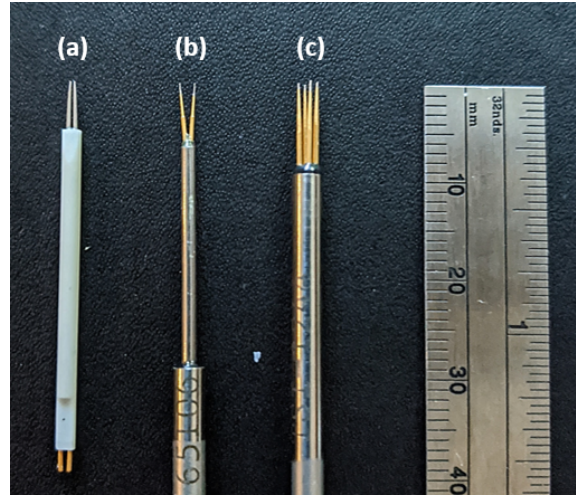


Figure 4: Probes used for temperature and velocity measurements (a) 55P31 (b) 1260A-T1.5 (c) 1244-T1.5

Prior to measurements, both temperature and velocity sensors are calibrated for the range of measurement conditions. The fine wire temperature probes are calibrated using distilled water baths maintained at temperatures between 283 K and 368 K. Reference temperature measurements are performed using an OMEGA K-type thermocouple placed approximately 0.25 inches from the sensor. The calibrations, verified for repeatability, yield a linear relationship between bridge voltage ( $E$ ) and jet total temperature ( $T_0$ ). Representative calibration data for a 55P31 probe is shown in fig. 5a. Hotwire sensors for velocity measurements are calibrated in situ in the jet potential core near the nozzle exit where turbulence intensities are low and flow velocities are obtained from the plenum pressure ( $P_0$ ) using isentropic relations. For velocity measurement in heated jets using multi-sensor probes, calibrations are performed at various jet stagnation temperatures ( $T_0$ ) between 295 K and 363 K to yield a family of curves, bridge voltage ( $E$ ) versus jet exit velocity ( $U_j$ ). Representative calibration data obtained for the velocity sensor of the 1244-T1.5 parallel wire probe is shown in fig. 5b. Velocity calibrations follow King's law, as shown in eq. (7). The constants A, B and n are recorded from the fitted curves.

$$E^2 = A + BU_j^n \quad (7)$$

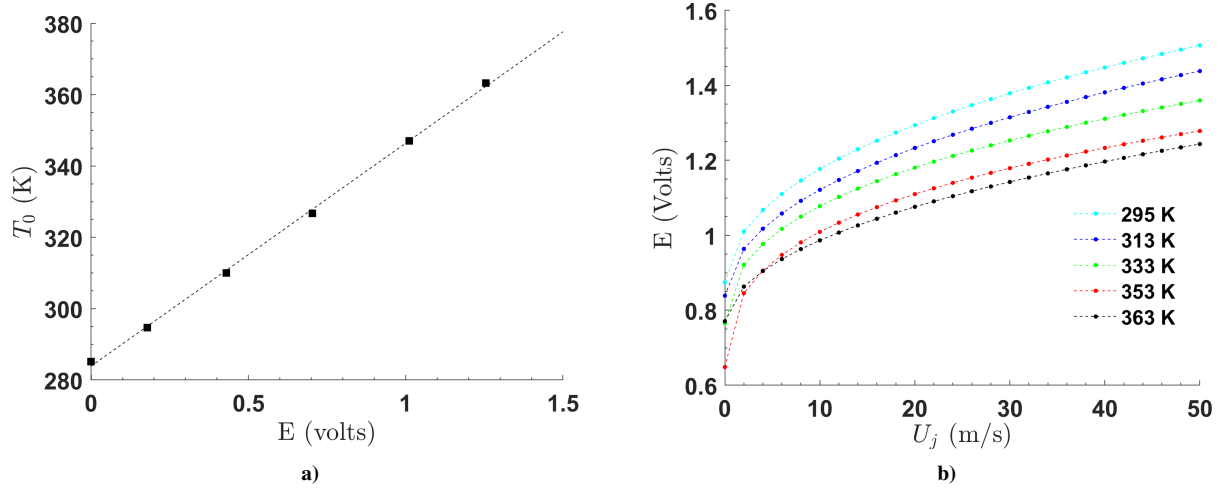


Figure 5: Representative calibration data showing, (a) temperature calibration curve for 55P31 sensor, (b) family of velocity calibration curves at various jet temperatures for the 1244-T1.5 velocity probe.

With the parallel wire probe, bridge voltages for the temperature and velocity sensors are simultaneously recorded at each measurement location. First, time series for temperature is evaluated using temperature calibration equation. The temperature data along with the bridge voltage time series for the velocity sensor is then used to interpolate velocity values from the family of calibration curves. A 3-axis computer controlled traversing system is used to perform velocity and temperature surveys in the axial and radial directions.

### III. Results

#### A. Mean velocity and temperature distribution

Mean velocity and temperature distributions for cold and heated jets are presented. Velocity measurements at the cold condition are from the single sensor probe (1260A-T1.5,  $d_w = 3.8 \mu m$ ). For the heated jet, velocity and temperature data are obtained simultaneously using the parallel wire probe (1244-T1.5,  $d_w = 3.8 \mu m$ ). It should be noted that mean temperature distributions obtained with the parallel probe are in good agreement with corresponding data taken with a single sensor probe (55P31,  $d_w = 1 \mu m$ ); the comparisons are not shown for brevity.

Figure 6 presents radial distribution of mean streamwise velocity ( $u$ ) near the nozzle exit at  $x/D = 0.040$  for  $T_0 = 295$  K and 353 K, respectively. Note that the survey only covers radial locations up to  $r/D \approx -0.35$ , where  $r/D = 0$  corresponds to the jet centerline. Velocity values are normalized with the isentropic jet exit velocity,  $U_j$ . As expected, the cold jet profile is characterized by rapid increases in velocity values across a thin shear layer followed by constant velocities in the jet core. At the heated condition, the increase from ambient to core velocities is more gradual, resulting in a thicker initial shear layer. Shear layer thickness,  $\delta$ , is estimated to be approximately  $0.020D$  for the cold and  $0.042D$  for the heated conditions. Here  $\delta$  is defined as the difference between radial locations ( $r$ ) where local velocities are 10% and 90% of  $U_j$ .

Corresponding mean temperature distribution for  $T_0 = 353$  K is presented in fig. 7. The y-axis represents normalized temperatures where,  $T$  is the time averaged temperature,  $T_a$  the mean ambient temperature, and  $\Delta T = T_0 - T_a$  is the difference between total temperature in the plenum and the ambient. Temperature values at the outer radial locations are observed to be higher than the ambient. A gradual increase in temperature occurs as one progresses inwards across the nozzle lip up to  $r/D \approx -0.52$ . For reference, the inner nozzle lip at  $r/D = -0.5$  and the lip thickness of 0.0625 in. are indicated by the two dotted vertical lines in fig. 7. The observed increase in temperature in the outer region of the profile is most likely due to local heating caused by radiation from the nozzle lip; note that at steady state the nozzle surface temperature approaches the total temperature in the plenum. The data is also noisier possibly because of low-frequency fluctuations associated with entrainment of ambient air. Following this, a rapid increase in temperature occurs across the shear layer eventually approaching  $T_0$  in the jet core. It is noted that temperature values in the core are slightly lower than  $T_0$  and is likely due to heat loss through the adapter and nozzle as the flow exits the nozzle.

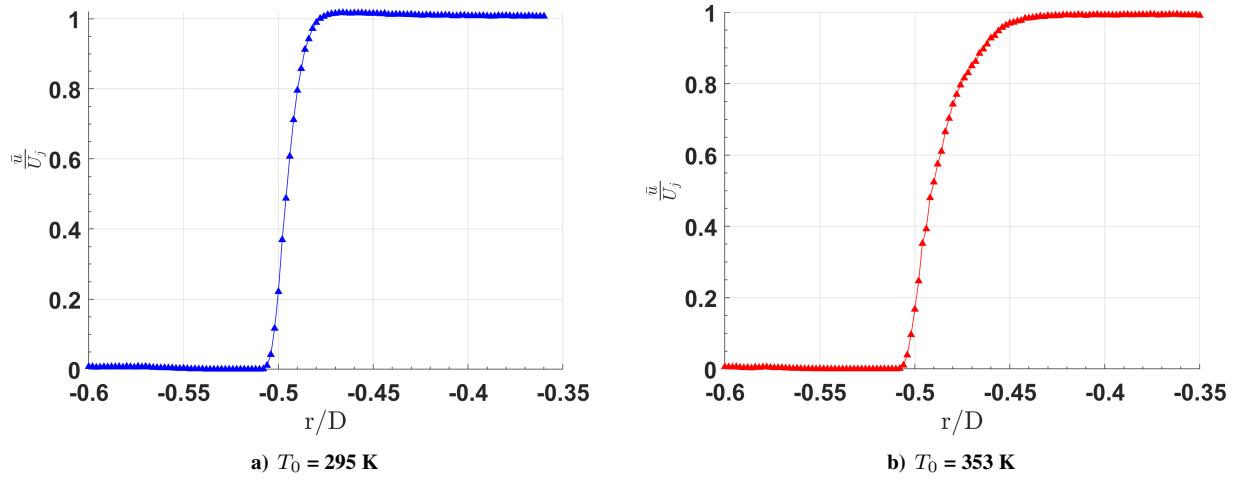


Figure 6: Mean axial velocity profiles obtained near the nozzle exit at  $x/D = 0.040$  for (a)  $T_0 = 295$  K (probe 1260A-T1.5) (b)  $T_0 = 353$  K (probe 1244-T.15);  $M = 0.08$ .

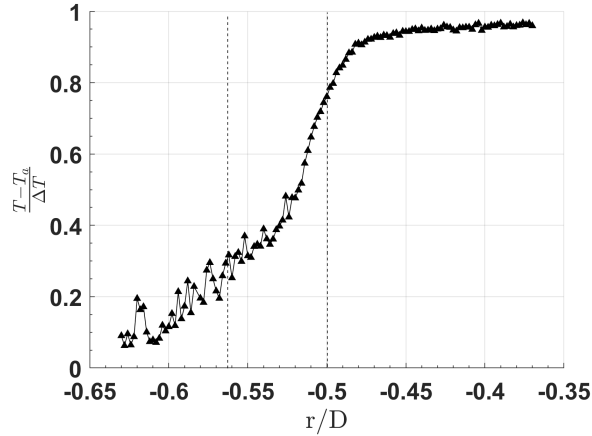


Figure 7: Mean temperature profile obtained near the nozzle exit at  $x/D = 0.040$  for  $T_0 = 353$  K (Probe 1244-T1.5);  $M = 0.08$ . The dotted lines represent the location of the nozzle lip.

Figure 8a presents the centerline distribution of mean axial velocity for the cold and heated cases. The profiles show a region of constant velocity near the nozzle exit followed by a rapid decay downstream of the potential core. Note that the inverse of mean velocity is plotted that clearly exhibits a linear asymptotic decay. The straight lines in the asymptotic regions of these curves are best fits with the slopes indicated in the legends. While the potential core for both cold and heated jets end at  $x/D \approx 6$ , the velocity for the heated jet decays at a faster rate downstream. This indicates increased jet mixing at heated conditions. Comparing these results to Witze's [26] correlation for centerline velocity decay, a reasonable agreement in case of the cold jet is obtained. The deviations, however, are notably larger for the heated case. The corresponding centerline temperature distribution for the heated jet is presented in fig. 8b. Qualitatively, temperature follows a trend similar to that of velocity. However, centerline temperature decays faster compared to velocity, as indicated by the shorter core length as well as a faster decay rate downstream (compare with fig. 8a red curve, noting the difference in ordinate scale).

Figure 9 presents radial distribution of mean streamwise velocity measured at various axial locations. The evolution of the cold and heated jet show similar trends, except the core values decay faster for the heated case, as also observed in fig. 8a. Heated jet profiles near the nozzle exit, for example at  $x/D = 0.5$  and 2, have a few anomalous data on the far end. These spurious points are likely attributable to probe interference effects when the probe support spans the width of the jet in order to reach the far side.

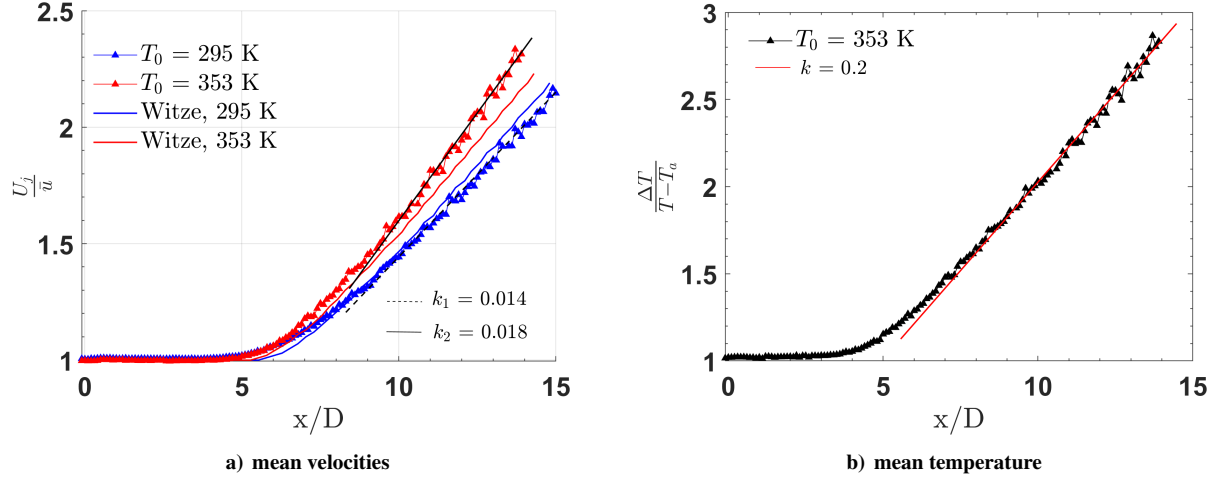


Figure 8: Centerline distribution of mean velocities and mean temperature (Probes 1260A-T1.5 for  $T_0 = 295$  K and 1244-T1.5 for  $T_0 = 353$  K);  $M = 0.08$ .

Radial distribution of mean temperature is illustrated in fig. 10. The temperature profiles are slightly wider compared to their velocity counterparts, indicating a thicker thermal mixing layer compared to the velocity shear layer (compare figs. 9b and 10). This is consistent with the faster decay of temperature in the jet centerline compared to velocity, as shown in fig. 8b. Finally, it is noted that temperature values in the outer shear layer on the right hand side are higher than the ambient. Recalling that these data are obtained with the parallel probe, this is most likely due to the temperature wire detecting the wake from the heated velocity wire as the entrained ambient air flows into the jet. Note that in the parallel probe, the velocity and temperature wires are 0.05 in. apart with the velocity wire located on the far side. As the mixing layer grows downstream, the onset of these spurious  $T$  data also shifts to larger radial locations. Conversely, on the near side (left in fig. 10), the wake of the ‘cold’ temperature wire flows over the heated velocity wire. Hence, no such spurious effect in the temperature data is observed.

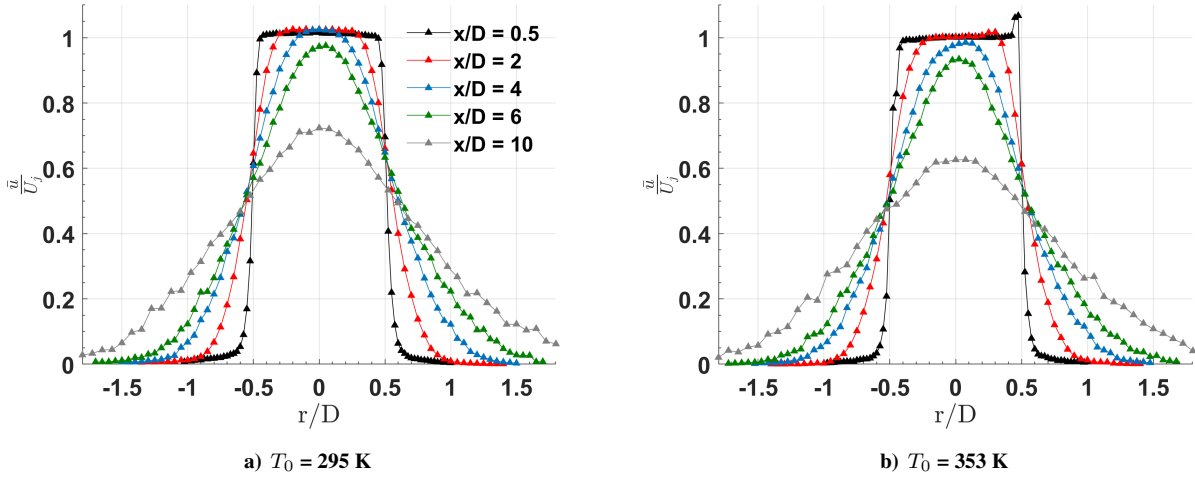


Figure 9: Radial distribution of mean streamwise velocity at various axial stations at (a) cold ( $T_0 = 295$  K, Probe 1260A-T1.5) (b) heated ( $T_0 = 353$  K, Probe 1244-T1.5) conditions;  $M = 0.08$ .

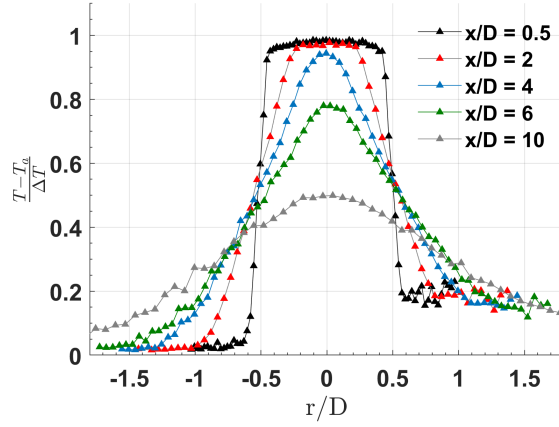


Figure 10: Radial distribution of mean temperature at various axial stations at heated ( $T_0 = 353$  K, Probe 1244-T1.5) condition;  $M = 0.08$ .

### B. Measurement of temperature and velocity fluctuations

Figure 11 shows centerline distribution of rms temperatures at  $T_0 = 353$  K. The corresponding plot for the mean temperature distribution can be referenced in fig. 8b. Repeat measurements are performed using a single wire (55P31,  $d_w = 1.3 \mu m$ ) and a parallel wire probe (1244-T1.5,  $d_w = 3.8 \mu m$ ). While both sensors capture similar trends, large differences in  $T_{rms}$  magnitudes due to different frequency responses of the wires are noted. Centerline  $T_{rms}$  distribution shows a region of constant, low temperature fluctuations in the core of the jet for  $x/D < 4$ . As the core dissipates, rms temperature fluctuations rapidly increase attaining peak values at  $x/D \approx 6$ . This is followed by a region of fast decay up to  $x/D \approx 9$  and a more gradual decay further downstream. Both sensors measure similar temperature fluctuations in the potential core region where the turbulence intensities are low. Downstream, significant differences occur; as expected, the finer wire ( $d_w = 1.3 \mu m$ ) measures higher magnitudes. For example, at  $x/D = 6$ , the peak  $T_{rms}$  magnitude measured by the  $1.3 \mu m$  wire is approximately 20% higher. Further downstream, the differences become progressively smaller, with both wires measuring similar levels towards the end of the measurement domain. When compared to previous measurements performed at GRC using Raman spectroscopy for a  $M = 0.38$  jet at  $T_0 = 520$  K, peak centerline  $T_{rms}$  values obtained with the  $1.3 \mu m$  wire are found to be approximately 21% lower [13, 27]. It remains unclear if the observed discrepancy is solely due to the low frequency response of the wire or in part because of differences in temperature, Mach number, and possibly initial boundary layer state of the two jets.

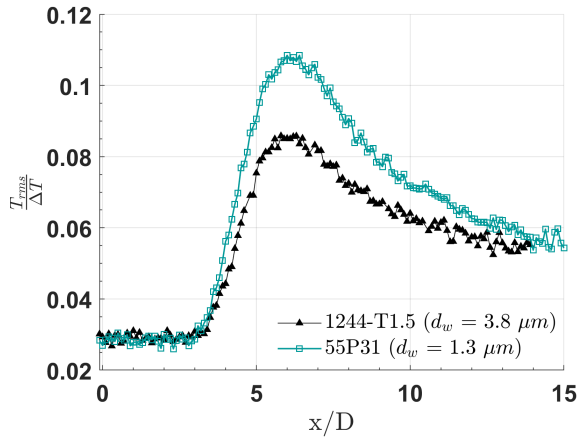


Figure 11: Centerline distribution of rms temperature for the heated jet ( $T_0 = 353$  K) measured using two different sensors;  $M = 0.08$ .

Figure 12 presents radial distribution of  $T_{rms}$  at various axial stations. Once again, qualitative trends captured by the two probes are in agreement with temperature fluctuations peaking in the mixing layer and declining in the jet core. At near nozzle locations, for example at  $x/D = 2$  and 4,  $T_{rms}$  distributions exhibit dual peaks in the shear layer; these peaks are captured by both the wires. As the jet evolves downstream, mixing with the ambient air results in thicker thermal mixing layer with progressively declining peak  $T_{rms}$  values. Temperature fluctuations captured by both sensors match fairly well in the jet core and outer shear layer regions where turbulent intensities are low. In regions of high shear, measured temperature fluctuations including the peak values are lower for the  $d_w = 3.8 \mu m$  wire. The differences between the two sensors, however, become progressively smaller as the jet evolves downstream, as also observed in the centerline data in fig. 11. Spurious  $T_{rms}$  values measured by the 1244-T1.5 parallel probe are noted in the outer shear layer on the jet far side (right hand side). This is due to interference from the wake of the heated velocity wire adjacent to the temperature wire, as discussed with fig. 10. Compared to previous Raman results [13], peak rms fluctuations at various axial stations obtained with the  $1.3 \mu m$  wire are about 20 – 25% lower. The dual peaks were not observed in the Raman results, likely due to inadequate spatial resolution in that measurement. We note here that LES results for the same jet, albeit with levels even higher than that measured by Raman, exhibit dual  $T_{rms}$  peaks in the mixing layer at  $x/D = 2$  and 4 [13, 27].

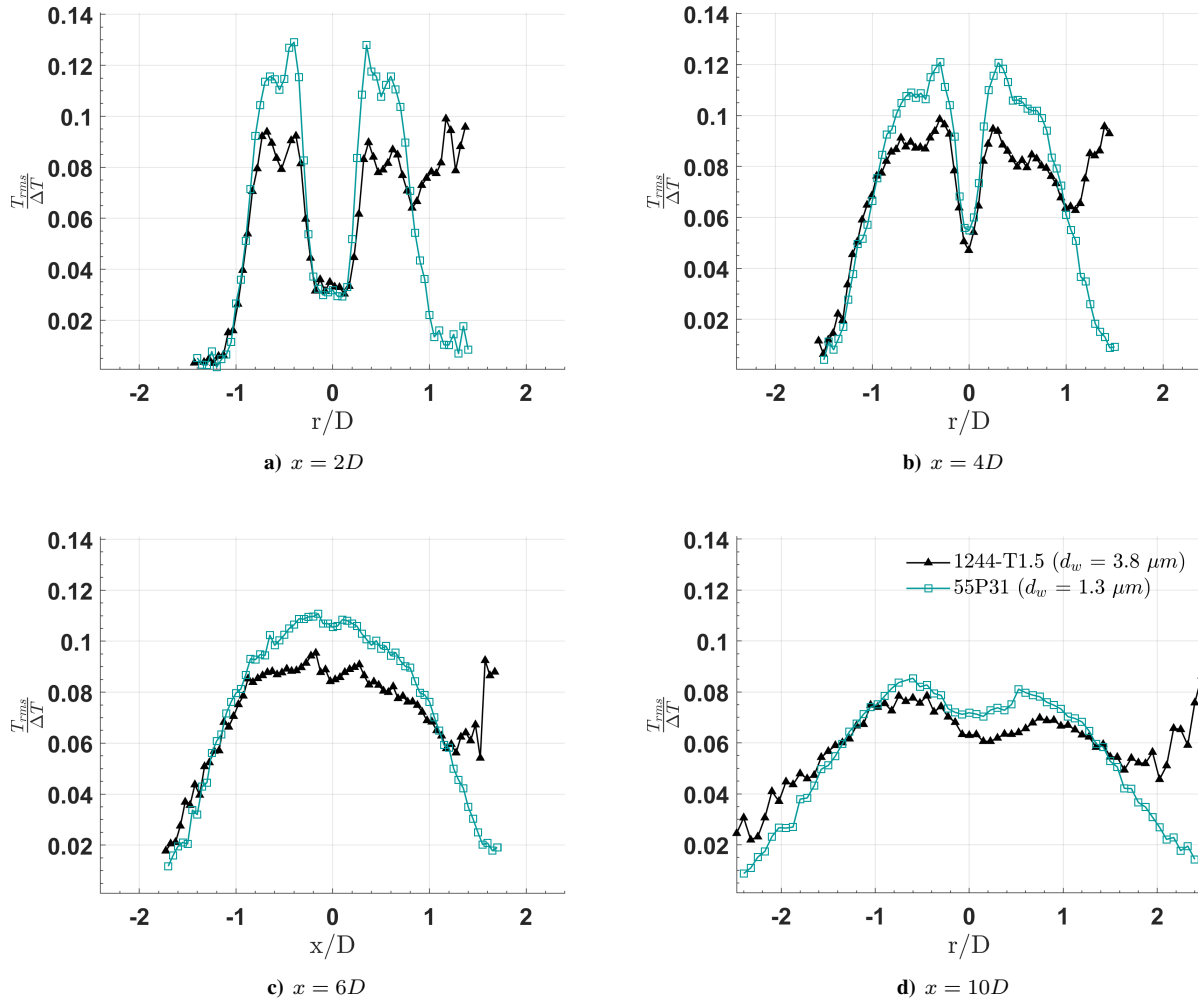


Figure 12: Radial distribution of rms temperature at various axial stations for the heated jet ( $T_0 = 353$  K);  $M = 0.08$ .

Here, a discussion is in order regarding the rms temperature measurement with the sensors of two different sizes. With reference to the discussion in section I, the frequency response of a fine wire sensor depends on its thermal mass as well as the end conduction effects. Moreover, the sensor cut-off frequency due to thermal inertia also increases with flow velocity as shown in fig. 3. For the two temperature sensors used in the current study, a platinum wire ( $d_w = 1.3 \mu m$ ,  $l_w = l_s = 0.4$  mm) and a platinum coated tungsten wire ( $d_w = 3.8 \mu m$ ,  $l_w = 1.5$  mm and  $l_s = 1.25$  mm), the theoretical cut-off frequencies at a mean velocity of 15 m/s are 3600 Hz and 690 Hz, respectively. In figs. 11 and 12, the

differences in measured  $T_{rms}$  magnitudes between the two wires are observed to be significant in regions of high shear where mixing at various frequencies and scales are anticipated. Moreover, these differences become progressively smaller as the jet evolves downstream. This is anticipated as the characteristic frequencies in a jet generally scale with the local shear layer thickness. Closer to the nozzle exit the shear layer is thin and turbulent fluctuations are expected to occur at high frequencies. In these regions, the  $3.8 \mu m$  wire experiences a lack of response resulting in underestimated  $T_{rms}$  magnitudes. As the shear layer thickens downstream, temperature fluctuations at lower frequencies dominate, and the differences in the measured amplitudes for the wires become smaller. Moreover, a reduced response at low to mid-range frequencies due to end conduction effects, as discussed in section I, is also expected to contribute to the overall discrepancy between the data from the two wires. Despite its better performance, it is important to understand that the frequency response of the  $1.3 \mu m$  wire may still not be adequate.

RMS velocity distributions are discussed next. Velocity measurements for the heated jet are performed with the parallel sensor probe ( $d_w = 3.8 \mu m$ ). As discussed in section II B, instantaneous bridge voltage values for the velocity sensor along with the simultaneously measured temperature data are used to evaluate velocity values at each measurement location. However, the  $3.8 \mu m$  wire in the 1244-T1.5 parallel probe clearly underestimates temperature fluctuations in the jet (figs. 11 and 12). Hence, the  $u_{rms}$  values evaluated using raw temperature data (uncompensated) are expected to have some error. Here, for the  $u_{rms}$  distributions presented below, the discussion will focus on the compensated  $u_{rms}$ ; uncompensated  $u_{rms}$  results are also included for comparison. The ‘compensated’  $u_{rms}$  data are obtained when the temperature time series data are first corrected for response lag using a frequency compensation technique. The compensated temperature time-series is then used to evaluate velocity data. Since it involves several steps, a discussion of the frequency compensation technique is deferred for later in section C.

The centerline rms velocity distributions are compared between the cold ( $T_0 = 295$  K) and the heated ( $T_0 = 353$  K) conditions in fig. 13. Only compensated  $u_{rms}$  values for the heated jet, shown in red, are discussed here. Comparison with uncompensated data will be discussed in section C. Higher turbulence intensities in the jet centerline near the nozzle exit are noted for the heated jet. The intensities for the heated case continue to rapidly increase resulting in a narrower peak that occurs a few diameters upstream relative to the cold case. This is followed by a faster decay of  $u_{rms}$  values downstream. Peak turbulence intensities are about 13% at  $x/D = 9.5$  and 16% at  $x/D = 7.5$  for the cold and heated jets, respectively. The trends observed in the cold jet data agree well with the literature [28, 29, 30, 31]. Peak turbulence intensity for the heated jet as well as the axial location at which it occurs agree reasonably well with heated subsonic jet data reported in the literature [32]. Comparative trends between the two temperature cases also match those observed by others [33]. Note that a ‘hump’ occurs for both cold and heated distributions in the  $x/D$  range between 1 and 4. This is presumably the signature of energetic large-scale coherent structures in the shear layer that might be more pronounced in the present low Mach number jet [28, 29]. Comparing the heated jet  $u_{rms}$  to the simultaneously measured  $T_{rms}$  data shown in fig. 11, it is noted that the rms temperature peaks slightly upstream of the rms velocity. This behavior was also noted by Locke and Wernet [13] with a higher Mach number and temperature jet. Also, unlike centerline temperature fluctuations that remain constant for the first 3-4 jet diameters,  $u_{rms}$  values appear to show a gradual increase for  $x/D > 1$ .

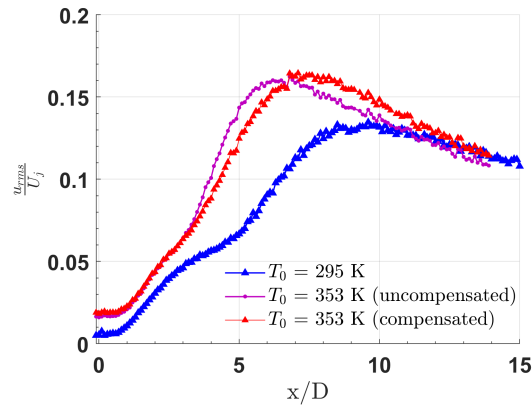


Figure 13: Centerline distribution of rms velocity for the cold ( $T_0 = 295$  K, Probe 1260A-T1.5) and heated ( $T_0 = 353$  K, Probe 1244-T1.5) jets;  $M = 0.08$ .

The radial distributions of  $u_{rms}$  at various axial stations for cold and heated jets are compared in fig. 14. Relative to the cold case, the heated jet exhibits a thicker mixing layer with increased velocity fluctuations observed on the high speed (inner) side of the jet. Also, the peak  $u_{rms}$  levels are higher and occur at inner radial locations (closer to the center). Profiles of  $u_{rms}$  for the heated jet can be compared to corresponding radial distribution of rms temperatures presented in fig. 12. As expected, temperature and velocity rms profiles show similar behavior except that the velocity fluctuations in the shear layer near the nozzle exit, for instance at  $x/D = 2$  and 4, do not exhibit dual peaks.

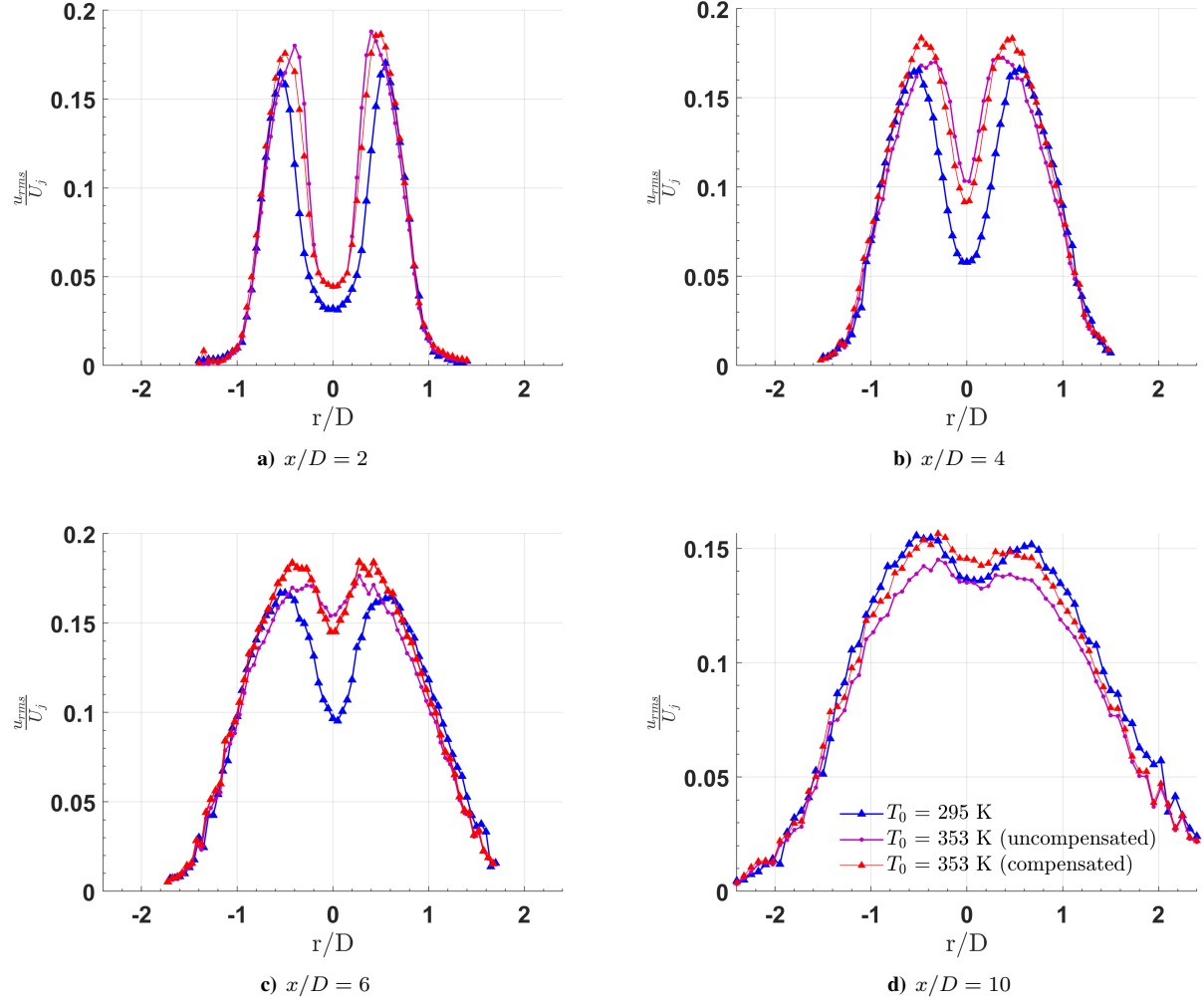


Figure 14: Radial profiles of rms velocity at various axial stations at cold ( $T_0 = 295$  K, Probe 1260A-T1.5) and heated ( $T_0 = 353$  K, Probe 1244-T1.5) conditions;  $M = 0.08$ .

### C. Frequency compensation and on-going efforts

Figure 15 compares temperature data acquired using two wires,  $d_w = 1.3$  and  $3.8 \mu m$ . Comparison is performed at two axial stations,  $x/D = 2$  and 10, for locations in the shear layer where peak  $T_{rms}$  levels are recorded. These correspond to radial locations of  $r/D = 0.4$  and  $0.65$  for  $x/D = 2$  and 10, respectively (fig. 12). Time series data in figs. 15a and 15c show the differences in temperature amplitudes between the two wires. Corresponding spectral plots in figs. 15b and 15d show the contribution of different frequencies on the measured  $T_{rms}$  levels. Theoretical cut-off frequencies ( $f_c$ ) of the wires based on local flow velocities are also listed in the corresponding plots. At  $x/D = 2$ , temperature spectra exhibit a prominent hump centered at  $f \approx 600$  Hz followed by spectral decay at higher frequencies. The peak frequency corresponds to a Strouhal number of  $St = fD/U_j \approx 0.5$  and is likely the signature of large-scale coherent structures [28, 29]. At  $x/D = 10$ , the spectral peak is relatively broad with most of the energy concentrated at low frequencies ( $f < 400$  Hz). At both locations, spectral amplitudes measured by the two wires are similar at low frequencies. The effect of wire frequency response become significant for  $f > 400$  Hz with differences in spectral levels progressively increasing at higher frequencies. At  $x/D = 2$ , peak energy is concentrated between 400-1000 Hz. For this reason, the  $3.8 \mu m$  wire with  $f_c \approx 700$  Hz underestimates  $T_{rms}$  amplitudes. The resulting effects on  $T_{rms}$  profiles are discussed in the following.

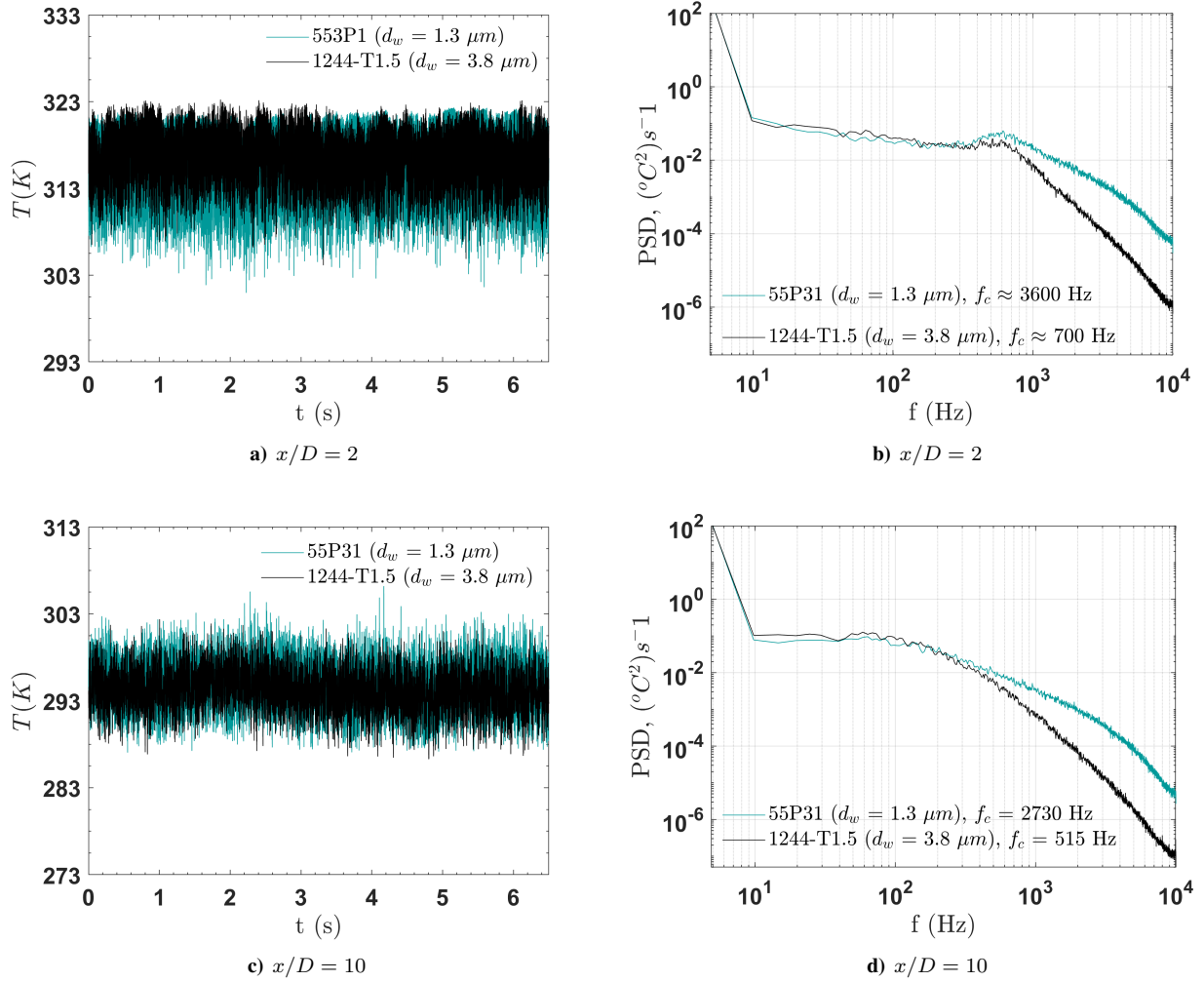


Figure 15: Comparison of temperature time series (left) and corresponding spectra (right) acquired using two wires,  $d_w = 1.3$  and  $3.8 \mu m$  at two axial stations (a, b)  $x/D = 2$  (c, d)  $x/D = 10$ . The data points correspond to location in the shear layer where peak  $T_{rms}$  values are measured.

The effect of frequency compensation on the temperature data is briefly investigated. As mentioned in section I, the compensation scheme outlined by Tagawa et al. [21] is followed. In short, this method estimates theoretical frequency response of fine wire sensors taking into account the effect of thermal inertia as well as end conduction losses. A time-dependent heat transfer equation for the wire and support structures (stubs, prongs) is solved to derive the response function. The interface between the wire and the support prong is treated with a boundary condition assuming no loss (i.e., rate of heat conducted from the wire is equal to the rate of heat conducted into the prongs). The result is a solution expressed by a transfer function in complex form; further details are outlined in the cited reference. Figure 16 shows representative amplitude and phase plots for the wire response function  $H(\omega)$  calculated for mean velocity and temperature conditions,  $u = 15$  m/s and  $T = 323$  K. The amplitude ( $|H(\omega)|$ ) curve clearly shows a first dip at low frequencies ( $f < 1$  Hz) due to end conduction from the wire to the prongs. A region of flat frequency response is then encountered for both the wires. The  $1.3 \mu m$  wire, owing to its lower thermal mass, exhibits a flat response over a larger frequency range. This is followed by attenuation at high frequencies due to thermal inertia of the wires. The theoretical -3dB cut-off frequencies of the wires are listed in the figure. The phase plot shows frequencies at which the wire's response lags temperature fluctuations in the flow. Once again, distinct dynamic response of the wire at low and high range of frequencies are noted. Frequency response plots clearly show that in addition to the effect of wire thermal inertia, losses due to end conduction at low to mid-range frequencies, where higher spectral energies are

encountered (see fig. 15), are also a significant contributor to total  $T_{rms}$  errors.

To apply frequency compensation, time series temperature data is converted to frequency domain using Fast Fourier Transform. The data is then multiplied point by point with the inverse of frequency response function ( $H(\omega)$ ). Inverse FFT on the compensated spectra is then performed to obtain time-series of the compensated temperature data. As an example, temperature compensation using this method is applied to data presented in fig. 15 and the results are shown in fig. 17. Compared to the raw data, compensated data for both wires exhibit increased spectral amplitudes at all frequencies. The differences between the raw and compensated spectra are fairly uniform up to the cut-off frequency of the wire. In this region, the signal is compensated for a constant loss in gain due to end conduction effects. For frequencies above  $f_c$ , the gain in spectral amplitude as a result of compensation increases with frequency; as expected, the increase is larger for the  $3.8 \mu m$  wire. Upon comparing the compensated spectra for the  $1.3$  and  $3.8 \mu m$  wire, one notes that the spectral amplitudes for the former is higher for  $f > 1000$  Hz. Similar effects were noted when comparable cold wire sensors were used for temperature measurements in a film cooling experiment [12].

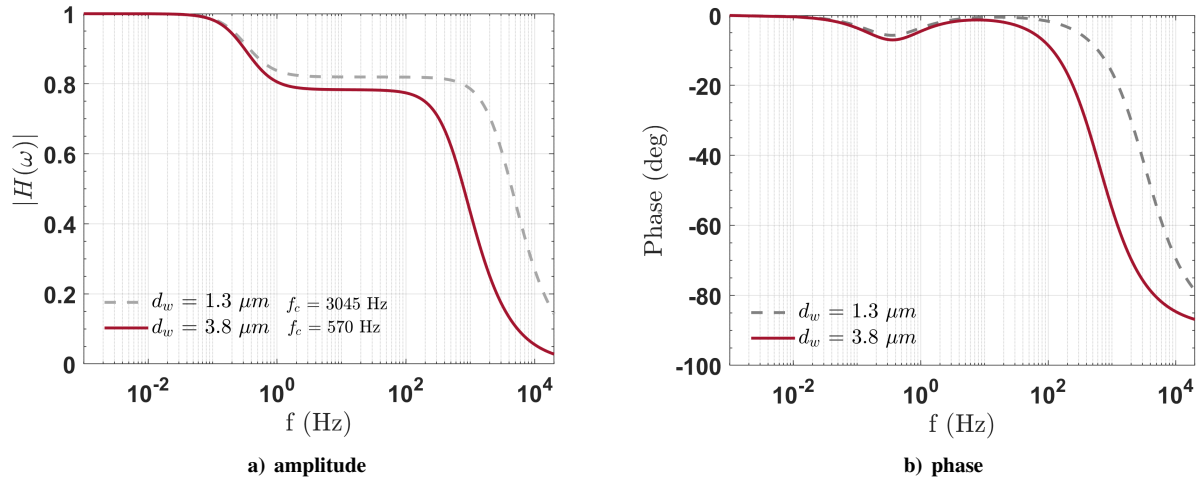


Figure 16: Theoretical frequency response solution for the two wires calculated at mean velocity ( $u = 15$  m/s) and temperature ( $T = 323$  K), showing (a) amplitude (b) phase.

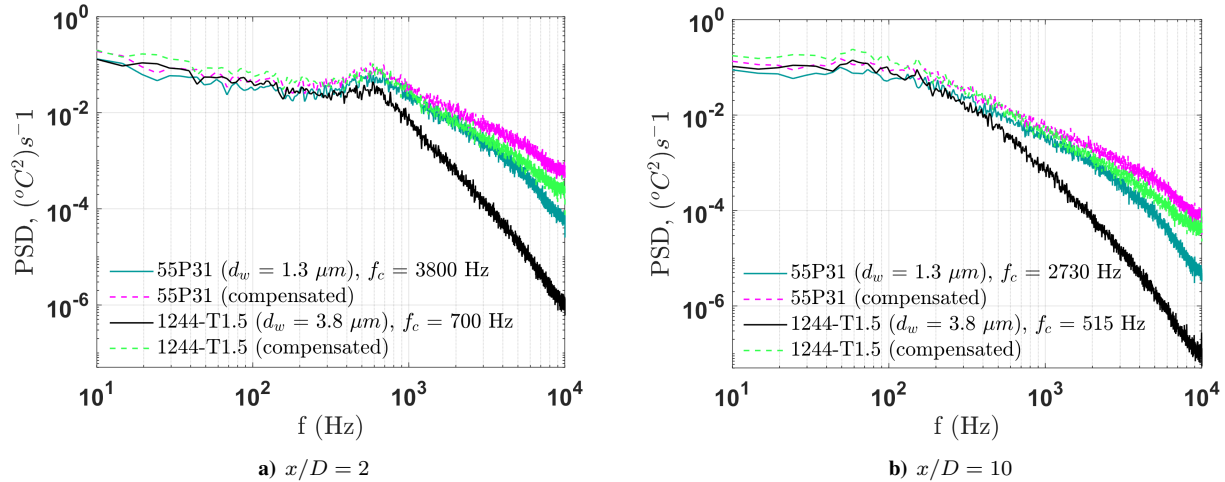


Figure 17: The effect of frequency compensation applied to the data of fig. 15 and the resulting temperature spectra at (a)  $x/D = 2$  (b)  $x/D = 10$ .

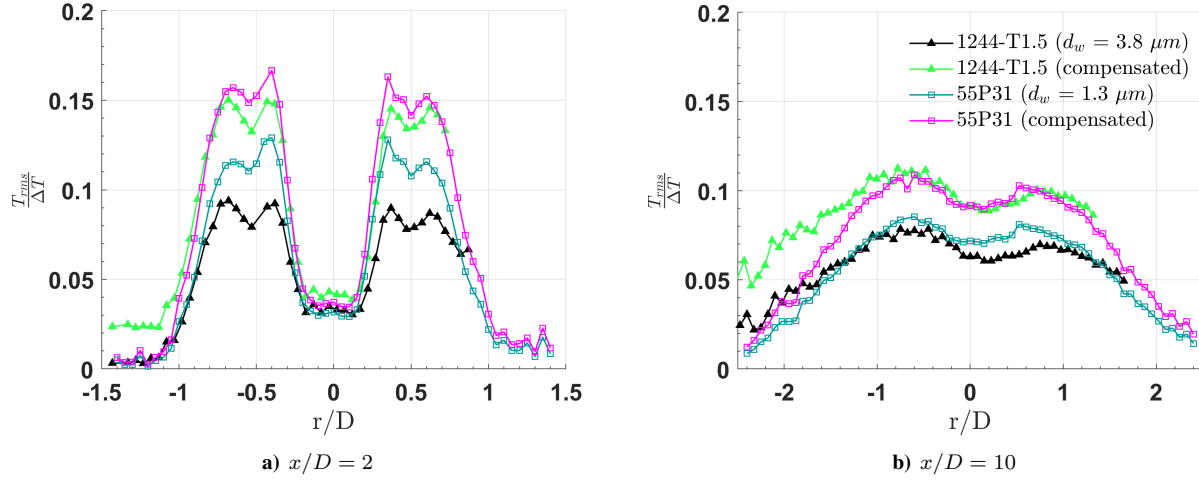


Figure 18: The effect of frequency compensation on radial distribution of  $T_{rms}$  at (a)  $x/D = 2$  (b)  $x/D = 10$ .

Overall effects of frequency compensation on the radial distribution of  $T_{rms}$  are presented in fig. 18. For both wires, compensation significantly increases the  $T_{rms}$  magnitudes in the shear layer. At  $x/D = 2$ , peak levels in the shear layer are mainly affected with only small changes in the jet core. Comparing the compensated results for the two wires, notable discrepancies are only observed in the shear layer where peak fluctuations for the  $1.3 \mu m$  wire are approximately 10% higher. Considering the large differences in raw signals between the two wires, the effect of compensation is encouraging and signifies that thicker wires may indeed be used for these measurements provided the compensation is done correctly. It is also noteworthy that with the compensated results for the  $1.3 \mu m$  wire, a much better agreement with the Raman data (discussed earlier [13, 27]) is obtained. Here, also recall that the centerline and radial profiles of compensated and uncompensated  $u_{rms}$  data for the heated jet were shown in figs. 13 and 14. In fig. 13, the difference in the amplitude of the peaks was not significant. However, curiously, the compensated  $u_{rms}$  distribution peaked approximately  $1D$  downstream compared to the uncompensated case. Similar shifts in the peak location due to compensation were also noted in the radial profiles. Perhaps this is due to steep gradients of the mean values of  $u$  and  $T$  around the locations of peak turbulent activity. This will be explored further in the future.

Finally, simultaneously acquired axial velocity and temperature data are used to calculate axial turbulent heat flux,  $\overline{u'T'}$ ; here,  $u' = u - \bar{u}$  and  $T' = T - \bar{T}$ , where  $u$  and  $T$  are instantaneous and  $\bar{u}$  and  $\bar{T}$  are time-averaged velocity and temperature, respectively. Figure 19 compares the heat flux values calculated using raw and compensated data.

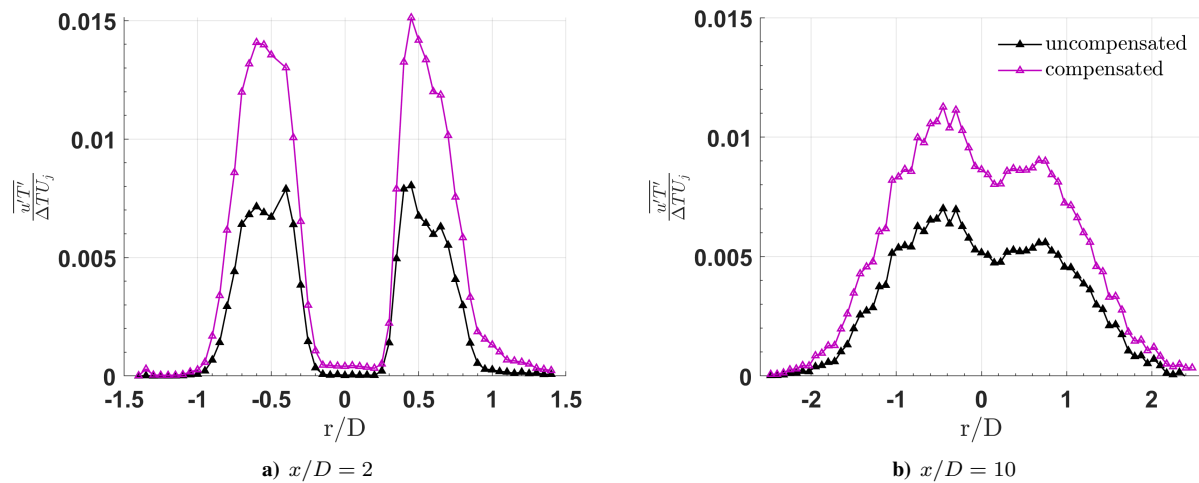


Figure 19: The effect of temperature compensation on the radial distribution of axial turbulent heat flux for (a)  $x/D = 2$  (b)  $x/D = 10$  (Probe 1244-T1.5);  $M = 0.08$ .

At  $x/D = 2$ , positive axial heat flux values are observed in the shear layer indicating turbulent transport of heat in the downstream direction. As expected, the effect of compensation is the most pronounced in regions of high shear. Peak  $\overline{u'T'}$  values are significantly enhanced; negligible effect occurs in the jet core and outer shear layer regions. At  $x/D = 10$ , the differences in raw and compensated heat flux values in the shear layer are comparatively smaller. When compared to the LES data for a heated jet at  $M = 0.38$  and  $T_0 = 520$  K, the peak heat flux for the compensated data at  $x/D = 2$  and 10 are observed to be about 20% and 10% lower, respectively [27].

In general, the results presented in this paper indicate that fine wire sensors with appropriate frequency compensation can be used adequately for measurement of time-resolved temperature in heated turbulent jets. Further refinement of the temperature post-processing technique, perhaps including input from the flow field, can possibly provide better compensation for the response delay of the wires. While reasonable comparisons with the literature are encouraging, a more direct comparison is planned for the near future with temperature and velocity measurements using a Rayleigh scattering technique in the same rig. It is hoped that a better understanding of the dynamic response for temperature measurements in heated mixing layers will be achieved.

## IV. Conclusion

Detailed time-resolved temperature and velocity measurements are carried out for a Mach 0.08 jet at total temperatures of 295 K and 353 K. Mean and rms temperature data acquired using two fine wire sensors, with diameters  $d_w = 1.3$  and  $3.8 \mu m$ , are compared. While both wires suffer from lack of frequency response due to thermal inertia and end conduction effects, the higher thermal mass of the  $3.8 \mu m$  wire results in largely underestimated  $T_{rms}$  values in the shear layer. In order to correct for the dynamic response of the wire, a post processing scheme based on the theoretical frequency response of the wires available in the literature is used. The results of frequency compensation are encouraging and demonstrate that the measurement bandwidth of the wires can be extended to include turbulent fluctuations at higher frequencies. Compensated temperature signals for the  $3.8 \mu m$  wire matched within 10% of the corresponding  $1.3 \mu m$  data. Considering the large differences in raw signals between the two wires, the effect of compensation looks promising. Corresponding velocity measurements are also performed at cold and heated conditions using single and parallel wire probes. Simultaneously measured temperature and velocity data are also used for axial heat flux calculations. The results are currently being further analyzed. In addition to a more rigorous comparison with the data available in the literature, direct comparisons of the current data will be performed in the future using Rayleigh Scattering technique in the same rig. Detailed time-resolved data presented in this paper provide an assessment of the efficacy of fine wire measurements; for the flow under consideration these measurements are considered adequate provided that a compensation for frequency response is performed. The results also provide further insight into the evolution characteristics, and serve to aid numerical predictions, of heated jets.

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